

Gallium

Ga (Ga I). Strongest classified transitions below 2 μm (NIST): 17 levels, 23 lines.

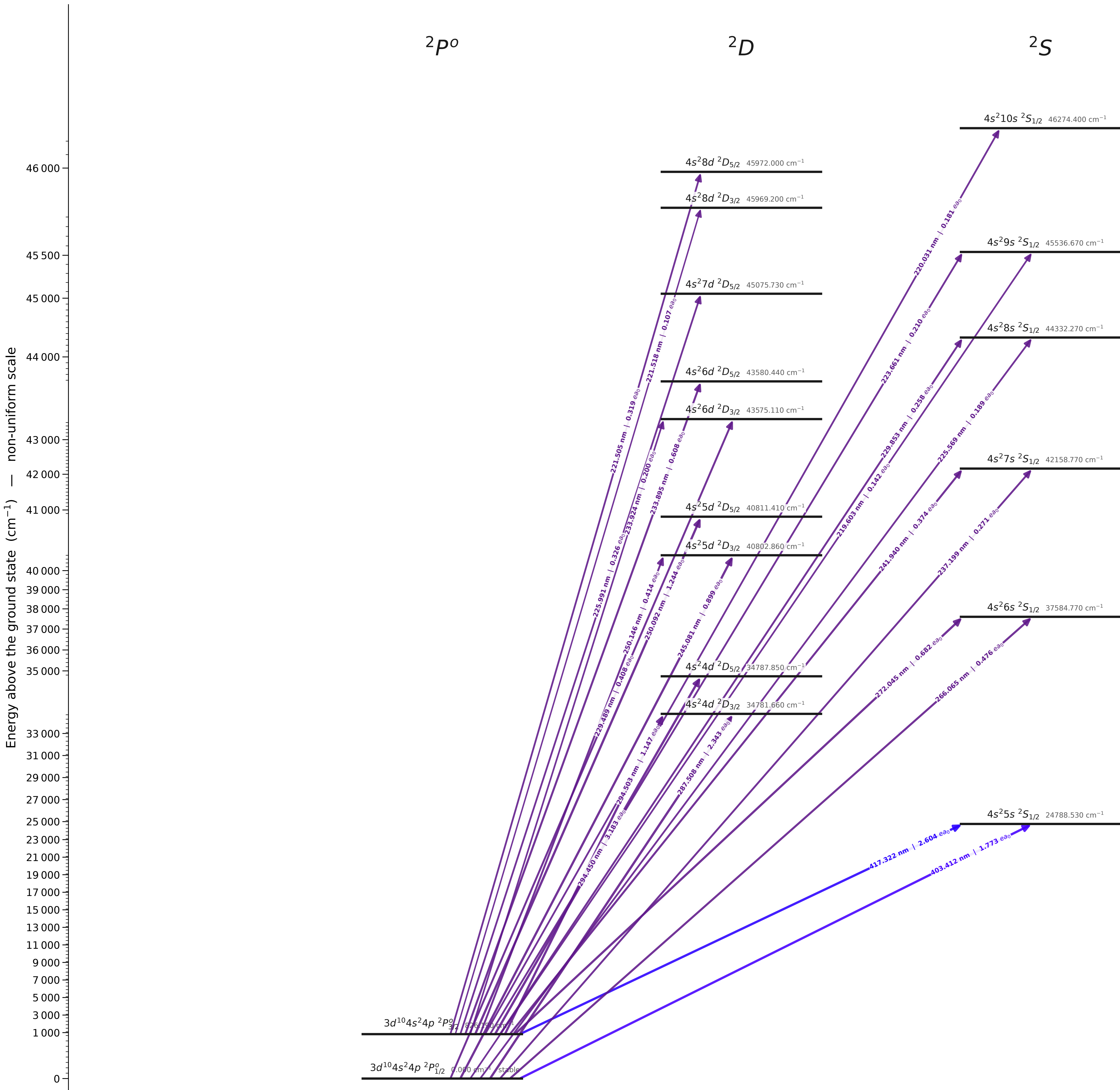
Ga⁺ ionisation limit 48387.63 cm⁻¹ = 5.99930 eV (206.664 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation

Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J er J' \rangle $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^29s\ ^2S_{1/2}$	223.6612	223.5918	1340.3865	0.210	4.000e+06	-	-	
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^26d\ ^2D_{3/2}$	229.4888	229.4181	1306.3489	0.408	6.970e+06	-	-	
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^28s\ ^2S_{1/2}$	229.8529	229.7821	1304.2795	0.258	5.560e+06	-	-	
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^26d\ ^2D_{5/2}$	233.8949	233.8232	1281.7402	0.608	9.750e+06	-	-	
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^27s\ ^2S_{1/2}$	237.1986	237.1262	1263.8881	0.271	5.570e+06	-	-	
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^27s\ ^2S_{1/2}$	241.9399	241.8664	1239.1196	0.374	1.000e+07	-	-	
$3d^{10}4s^24p\ ^2P_{1/2}^o - 4s^25d\ ^2D_{3/2}$	245.0809	245.0066	1223.2390	0.899	2.780e+07	-	-	
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^25d\ ^2D_{5/2}$	250.0924	250.0170	1198.7267	1.244	3.340e+07	-	-	
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^25d\ ^2D_{3/2}$	250.1459	250.0705	1198.4704	0.414	5.540e+06	-	-	
$3d^{10}4s^24p\ ^2P_{1/2}^o - 4s^26s\ ^2S_{1/2}$	266.0652	265.9861	1126.7631	0.476	1.220e+07	-	-	
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^26s\ ^2S_{1/2}$	272.0453	271.9647	1101.9945	0.682	2.340e+07	-	-	
$3d^{10}4s^24p\ ^2P_{1/2}^o - 4s^24d\ ^2D_{3/2}$	287.5078	287.4235	1042.7279	2.343	1.170e+08	-	-	
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^24d\ ^2D_{5/2}$	294.4497	294.3636	1018.1449	3.183	1.340e+08	-	-	
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^24d\ ^2D_{3/2}$	294.5034	294.4173	1017.9594	1.147	2.610e+07	-	-	
$3d^{10}4s^24p\ ^2P_{1/2}^o - 4s^25s\ ^2S_{1/2}$	403.4124	403.2984	743.1414	1.773	4.850e+07	-	-	
$3d^{10}4s^24p\ ^2P_{3/2}^o - 4s^25s\ ^2S_{1/2}$	417.3215	417.2039	718.3729	2.604	9.450e+07	-	-	

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.